

CPT208 Course QA Agent System

Cloud-based Student Support & Knowledge Management Platform

A cloud-based QA system that helps students find coursework information and helps teachers manage course knowledge efficiently.

Why This System? The Problems We Aim to Solve

01. The Critical Problems



Repeated Student Questions

Educators waste valuable time answering identical queries daily, leading to inefficiency and reduced time for personalized teaching.



Scattered Information

Key course details are siloed across PDFs, emails, and LMS posts, creating confusion and making accurate information hard to find.



Manual Scalability Limits

As student numbers grow, providing consistent, high-quality human support becomes logistically unmanageable and costly.

02. Our Strategic Goals

Instant Answers

Deliver responses to student inquiries, eliminating wait times and frustration for learners.

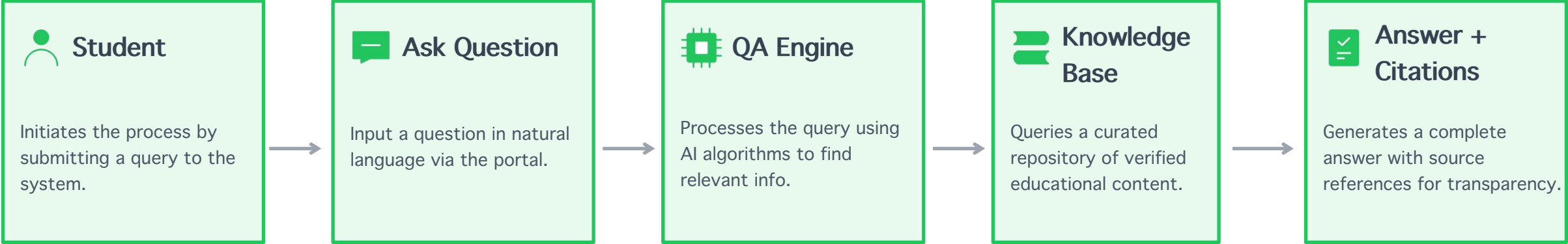
Reliable Citations

Ensure every answer links back to verified course materials, maintaining academic integrity and trust.

Knowledge Base Management

Empower educators to easily update and curate the system's core knowledge with minimal effort.

How It Works: A Deep Dive into System Features



Student Portal

Ask questions, view detailed answer citations, and access personal login & query history for continuous learning.

Teacher Portal

Manage QA workflows, review student inquiries, and create or update new knowledge base entries to enrich content.

Analytics Dashboard

Monitor popular questions, track system usage statistics, and oversee pending reviews for data-driven improvements.

Current Implementation: Feature List



Student Portal

Secure login, question submission, answer & citation viewing, query history.



Teacher Portal

Secure admin access, review queue for student inquiries and low-confidence answers.



Knowledge Base

Structured repository for course materials, supporting various document formats.



Knowledge Studio

Intuitive tools to create, edit, categorize, and maintain knowledge base content.



Extractive QA Engine

Generates precise answers with direct citations to source documents.



Analytics Dashboard

Visualizes FAQs, system usage patterns, and pending review status.



Confidence Scoring

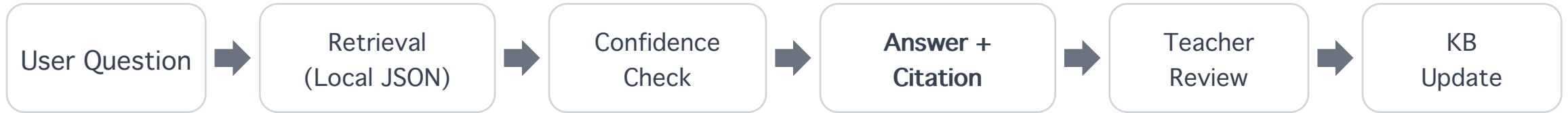
Automatically assesses answer reliability; flags low-confidence inquiries for teacher review.



Data Export Functionality

Supports exporting student queries and system analytics into CSV format.

Data Flow & Technical Architecture



Technical Stack

Frontend:Broad compatibility, Fast load

API Layer:Secure, Efficient communication

Backend:Node.js (lightweight, scalable)

Database:Supabase PostgreSQL (Managed)

Current Implementation

Extractive QA Mode using local JSON files. Operates without external APIs, ensuring data security and minimizing hallucinations.

Future Enhancement

Integration with OpenAI API for advanced text generation and Pinecone Vector DB for efficient semantic retrieval.

"The system is already deployed online and accessible from different devices."

Beyond CPT208: A Vision for Future Development



Current Prototype

Stable foundation for
CPT208



OpenAI Integration

Enhance feedback, Q&A, and
content generation



Pinecone Retrieval

Efficient, context-aware
knowledge retrieval



Multi-course Platform

Unified, flexible learning
ecosystem

Future Support: CPT208, CPT202, CPT204, and other computing courses.

"The system is designed as a reusable course-support platform rather than a CPT208-only application."

Evaluation & Evidence



Response Time

< 2 seconds

Average time to generate an answer with accurate citations from the knowledge base.



Answer Hit Rate

~90%

Percentage of student questions that receive a direct, relevant answer from the system.



Low-Confidence Queries

~10%

Inquiries flagged for teacher review due to insufficient confidence score.



Teacher Review Actions

~5%

Reviewed answers that required teacher correction or knowledge base update.



Top 3 FAQ Topics

Identified via popular question analytics

01. Assignment Deadlines

Most frequent time-related query

02. Project Requirements

Clarifications on deliverables

03. Submission Format

File type and structure help

Thank You

& Live Demo

<https://cpt208-course-desk-production-8daa.up.railway.app/>